

Preconditioned WENO finite-difference lattice Boltzmann method for simulation of incompressible turbulent flows

Kazem Hejranfar^{*}, Mohammad Hossein Saadat

Aerospace Engineering Department, Sharif University of Technology, Iran

ARTICLE INFO

Article history:

Received 29 May 2017

Received in revised form 31 March 2018

Accepted 23 June 2018

Available online 13 July 2018

Keywords:

Lattice Boltzmann method

Incompressible turbulent flows

WENO finite-difference scheme

Turbulence models

ABSTRACT

In this work, a preconditioned high-order weighted essentially non-oscillatory (WENO) finite-difference lattice Boltzmann method (WENO-LBM) is applied to deal with the incompressible turbulent flows. Two different turbulence models namely, the Spalart–Allmaras (SA) and $k - \omega$ SST models are used and applied in the solution method for this aim. The spatial derivatives of the two-dimensional (2D) preconditioned LB equation in the generalized curvilinear coordinates are discretized by using the fifth-order WENO finite-difference scheme and an implicit–explicit Runge–Kutta scheme is adopted for the time discretization. For the convective and diffusive terms of the turbulence transport equations, the third-order WENO and second-order central finite-difference schemes are used, respectively. The preconditioning technique along with the local time-stepping method are applied to the WENO-LBM to further accelerate the solution to the converged steady-state condition, and therefore, an accurate and efficient incompressible LB solver is provided for simulating turbulent flows. The accuracy and robustness of the proposed solution method are assessed by computing two test cases: the 2D turbulent flow over a flat plate at $Re = 1.0 \times 10^7$ and the 2D turbulent flow over a NACA0012 airfoil at $Re = 6.0 \times 10^6$ and different angles of attack. The present results are compared with the available numerical and experimental results which show excellent agreement. It is demonstrated that the present solution methodology based on the WENO-LBM provides more accurate results of the incompressible turbulent flows and it requires lower number of grid points compared to the traditional (low-order accurate) LB and Navier–Stokes solvers.

© 2018 Elsevier Ltd. All rights reserved.

1. Introduction

The Lattice Boltzmann Method (LBM) has gained increased popularity over recent years to simulate complex fluid flow problems. The LBM is based on the statistical mechanics and the kinetic theory in which the macroscopic flow variables are recovered from the dynamic evolution of the particle distribution functions in the phase space. The standard (conventional) LBM which consists of the streaming and collision steps has some advantages (e.g. ease of programming, inherent parallelism), however, it has some serious limitations too, due to the strong coupling of the time and space discretization. The standard LBM is restricted to uniform Cartesian grids with equal spacing which greatly limits its application in engineering problems; it is only second-order accurate in space and time and the most important one is its instability at high Reynolds number flows. He and Luo [1] demonstrated that the discretization of time and space does not necessarily need to be coupled, therefore, one can overcome the above mentioned limitations by adopting different numerical schemes such as finite difference [2–7], finite volume [8,9], finite element [10–12], discontinuous Galerkin spectral

^{*} Corresponding author.

E-mail address: khejran@sharif.edu (K. Hejranfar).

element [13,14], Chebyshev collocation spectral [15,16] and spectral difference [17] methods to directly discretize the lattice Boltzmann equation (LBE). From a computational point of view, the LB equation has some advantages over the Navier–Stokes equations for the fluid flow simulations. The LB equation is a simple linear hyperbolic equation which consists of only first-order derivatives and the pressure is calculated locally via an equation of state. Therefore, the numerical difficulty in the discretization of the Navier–Stokes equations due to the nonlinear convective terms and the second-order derivative diffusive terms and also the difficulty of the pressure–velocity coupling in the incompressible Navier–Stokes solvers are not addressed in the LBM. For precisely predicting fluid flows with complex phenomena such as turbulent flows, the use of high-order accurate numerical methods is of prime interest and since the LB equation has a simpler formulation than the Navier–Stokes equations, the implementation of high-order accurate finite-difference methods to solve the LB equation seems to be easier and consequently high-order accurate finite-difference LB solvers can be considered as suitable alternative to the high-order accurate finite-difference Navier–Stokes solvers.

In spite of the advantages of the LB equation over the Navier–Stokes equations, it has two important limitations. The first one is that the LB equation with the Bhatnagar–Gross–Krook (BGK) approximation becomes stiff at high Reynolds number turbulent flows and requires a very small time step, $\Delta t \approx O(\tau)$, to maintain the numerical stability [18]. One of the simple and effective ways to treat this problem is to use the implicit–explicit (IMEX) Runge–Kutta scheme for the time discretization [2]. In this method, the collision term which is the source of instability and stiffness is treated implicitly and the convective terms are treated explicitly. In this way, the restriction of $\Delta t \approx O(\tau)$ can be relaxed, however, the explicit discretization of the convective terms imposes a stability criterion on Δt through the Courant–Friedrichs–Lewy (CFL) condition, i.e. $CFL = \frac{\Delta t}{\min(\Delta x, \Delta y)} < 1$. Another important limitation of the LB equation is its slow convergence rate and poor computational performance for steady-state computations [19,20]. To overcome this shortcoming, one can apply a preconditioning technique [19,20] along with the local time step method to accelerate the steady-state solution. The main feature of this technique is that it retains the simplicity of the LB formulation and only by a slight modification to the LB equation, it is possible to considerably reduce the computational time to obtain the steady-state solution.

Several studies have been performed in the recent years on simulation of turbulence in the context of the LBM. However, most of them are based on the direct numerical simulation (DNS) [21–26] and the large eddy simulation (LES) [27–33] in simple geometries and the implementation of the traditional turbulence models to the LBM has now gained much attention because of their simplicity of implementation. Teixeira [34] applied an algebraic mixing length and $k - \varepsilon$ turbulence models to the single relaxation time (SRT) standard LBM and simulated the turbulent flow in a straight pipe and over a backward facing step. Filippova et al. [35] simulated the two-dimensional (2D) turbulent flow in an axial compressor cascades using the multi-scale LBM coupled with the $k - \varepsilon$ turbulence model. Chen et al. [36] combined the $k - \varepsilon$ turbulence model with the SRT LBM in the PowerFLOW code and simulated the turbulent flow over the NACA 4412 airfoil and a real car geometry. Shu et al. [37] developed the Taylor series expansion and least-squares-based LBM and combined it with the $k - \omega$ and SA turbulence models to simulate turbulent channel flow and turbulent flow over a backward facing step. Li et al. [38] incorporated the SA turbulence model into the multi relaxation time (MRT) LBM and simulated the turbulent flow over a NACA0012 airfoil at $Re = 5 \times 10^5$ and over a flat plate at $Re = 2 \times 10^4$. Sajjadi et al. [39] simulated the 2D particle transport and deposition by coupling the LBM with the standard and renormalization group $k - \varepsilon$ turbulence models. Shu and Yang [40] coupled the standard $k - \varepsilon$ model with the MRT LBM for the 2D simulation of the turbulent flow over the backward-facing step at $Re = 1.0 \times 10^7$. Then, they extended the solver to 3D and with the help of GPU programming they simulated the crude oil mixing in an industrial tank. The more recent study is the work of Guzel et al. [41] in which they applied the SA turbulence model to the second-order finite volume implicit–explicit LBM and simulated the turbulent flow over a NACA0012 airfoil at $Re = 6 \times 10^6$ and over a flat plate at $Re = 1 \times 10^7$.

Recently, we have proposed and applied the fifth-order WENO finite-difference method to the lattice Boltzmann equation (WENO-LBM) to accurately solve laminar incompressible flows in the generalized curvilinear coordinates with the non-uniform grids [5]. The main objective of the present study is to extend and apply this high-order accurate LB method for efficiently simulating incompressible turbulent flows. This aim is performed by applying the preconditioning technique to the WENO-LBM and implementing two turbulence models namely, the Spalart–Allmaras (SA) and $k - \omega$ SST models to the solution algorithm. The preconditioning technique along with the local time-stepping method applied to the WENO-LBM can improve the convergence rate of the solution to the steady-state condition, and therefore, the computation of incompressible turbulent flows can be accurately and efficiently performed. Note that the upwind nature of the WENO scheme guarantees that the scheme can readily be used in applications involving high Reynolds number turbulent flows. Herein, the convective and diffusive terms of the turbulence models in the computational domain are discretized by using the third-order WENO and second-order central schemes, respectively; and the first-order explicit method is applied for the time discretization. The accuracy and robustness of the present high-order accurate LB-based flow solver are examined by solving two incompressible turbulent flow problems: the 2D turbulent flow over a flat plate at $Re = 1 \times 10^7$ and the 2D turbulent flow over a NACA0012 airfoil at $Re = 6 \times 10^6$ and the angles of attack $\alpha = 0^\circ, 10^\circ, 15^\circ$. A sensitivity study is also performed to examine the effects of the preconditioning parameter on the accuracy and convergence rate of the steady-state solution. Results obtained by applying the present solution method are compared with the existing numerical and experimental results.

The paper is organized as follows: In Section 2, the LB equation in the generalized curvilinear coordinates with the preconditioning method for simulating the 2D incompressible flows are presented. Section 3 gives the turbulence models and their transformation into the computational plane. In Section 4, the discretization procedure based on the fifth- and third-order WENO finite-difference schemes is described and Section 5 presents the implementation of boundary conditions. Section 6 provides the results obtained by applying the WENO-LBM and finally some conclusions are given in Section 7.

2. Governing equations

The governing equation of the preconditioned lattice Boltzmann method is the same as the continuous Boltzmann equation as follows:

$$\frac{\partial f}{\partial t} + \mathbf{e} \cdot \nabla f = \Omega \quad (1)$$

in which f is the particle distribution function, $\mathbf{e}$ is the particle velocity and Ω is the collision term between particles. According to the Bhatnagar–Gross–Krook (BGK) approximation, the collision between particles drives the particle distribution function toward an equilibrium distribution f^{eq} with a single relaxation time τ [42]

$$\Omega = -\frac{1}{\tau}(f - f^{eq}) \quad (2)$$

A 2D square lattice model with nine velocity directions (D2Q9) is employed to discretize Eq. (1) in the lattice configuration. Therefore, the lattice Boltzmann (LB) equation for the particle distribution function f_α in the direction of the microscopic velocity $\mathbf{e}_\alpha$ may be written as

$$\frac{\partial f_\alpha}{\partial t} + \mathbf{e}_\alpha \cdot \nabla f_\alpha = -\frac{1}{\tau}(f_\alpha - f_\alpha^{eq}), \quad \alpha = 0, 1, \dots, 8 \quad (3)$$

where the subscript α denotes the direction of the particle speed. In the D2Q9 discrete Boltzmann model, the microscopic velocities are given as

$$\mathbf{e}_\alpha = (e_{\alpha x}, e_{\alpha y}) = \begin{cases} (0, 0) & \alpha = 0 \\ \left(\cos\left(\frac{\alpha-1}{2}\pi\right), \sin\left(\frac{\alpha-1}{2}\pi\right) \right) & \alpha = 1, 2, 3, 4 \\ \left(\cos\left(\frac{\alpha-5}{2}\pi + \frac{1}{4}\pi\right), \sin\left(\frac{\alpha-5}{2}\pi + \frac{1}{4}\pi\right) \right) \sqrt{2} & \alpha = 5, 6, 7, 8 \end{cases} \quad (4)$$

The equation of the equilibrium distribution function f_α^{eq} in the 2D preconditioned LBM is defined as [43,19,20]

$$f_\alpha^{eq} = w_\alpha \left(p + p_0 \left(3 \frac{\mathbf{e}_\alpha \cdot \mathbf{u}}{c_s^2} + \frac{9}{2} \frac{(\mathbf{e}_\alpha \cdot \mathbf{u})^2}{\gamma c_s^4} - \frac{3}{2} \frac{|\mathbf{u}|^2}{\gamma c_s^2} \right) \right) \quad (5)$$

where $0 < \gamma \leq 1$ is the preconditioning parameter, $\mathbf{u} = (u, v)$ is the velocity vector in the Cartesian coordinates, $c_s = 1/\sqrt{3}$ is the speed of sound and the weight coefficient w_α for the D2Q9 model is given by

$$\begin{aligned} w_0 &= 4/9, \\ w_1 &= \dots = w_4 = 1/9, \\ w_5 &= \dots = w_8 = 1/36. \end{aligned} \quad (6)$$

By applying the Chapman–Enskog expansion to the LB equation (3), the preconditioned incompressible Navier–Stokes equations can be derived as follows [43,19,20]:

$$\frac{1}{c_s^2} \frac{\partial p}{\partial t} + \nabla \cdot \mathbf{u} = 0 \quad (7)$$

$$\frac{\partial \mathbf{u}}{\partial t} + \frac{1}{\gamma} \mathbf{u} \cdot \nabla \mathbf{u} = -\frac{1}{\gamma} \nabla p^* + \frac{1}{\gamma} \nu \nabla^2 \mathbf{u} \quad (8)$$

where $p^* = \gamma c_s^2 \rho$ and the effective speed of sound is $c_s^* = \sqrt{\frac{\partial p^*}{\partial \rho}} = \sqrt{\gamma} c_s$. Eqs. (7) and (8) can be interpreted as the preconditioned Navier–Stokes equations as follows:

$$\frac{\partial Q}{\partial t} + PA \frac{\partial Q}{\partial x} + PB \frac{\partial Q}{\partial y} = PR \{Q_{vis}\} \quad (9)$$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\gamma & 0 \\ 0 & 0 & 1/\gamma \end{bmatrix}, A = \begin{bmatrix} 0 & 1 & 0 \\ \gamma c_s^2 & u & 0 \\ 0 & 0 & u \end{bmatrix}, B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & v & 0 \\ \gamma c_s^2 & 0 & v \end{bmatrix} \quad (10)$$

where $Q = \{p/c_s^2, u, v\}$ is the vector of primitive variables, P is the preconditioning matrix and $R\{Q_{vis}\}$ represents the appropriate viscous terms on the right hand side of Eq. (9). In this way, the eigenvalues of the preconditioned convective matrix PA becomes [19]:

$$\lambda = \left(u, u \pm u \sqrt{1 - \gamma + (\gamma c_s/u)^2} \right) \quad (11)$$

Therefore, by adjusting the preconditioned parameter γ , the disparity between the eigenvalues is reduced, and thus, the convergence rate of the solution of the LB equation can be increased [19]. When $\gamma = 1$ no preconditioning is performed.

The macroscopic fluid pressure p and the macroscopic velocity $\mathbf{u}$ are obtained from the following relations:

$$p = \sum_{\alpha} f_{\alpha}, \quad p_0 \mathbf{u} = \sum_{\alpha} \mathbf{e}_{\alpha} f_{\alpha} \quad (12)$$

where $p_0 = \gamma c_s^2 \rho_0$ and ρ_0 is the constant density of the fluid. The kinematic viscosity of the fluid ν and the relation for the relaxation time τ in the preconditioned finite-difference LB model is given by [44]

$$\tau = \frac{\nu}{\gamma c_s^2} \quad (13)$$

By using the chain rule, Eq. (3) in the generalized curvilinear coordinates can be written as

$$\frac{\partial f_{\alpha}}{\partial t} + \left(\tilde{e}_{\alpha\xi} \frac{\partial f_{\alpha}}{\partial \xi} + \tilde{e}_{\alpha\eta} \frac{\partial f_{\alpha}}{\partial \eta} \right) = -\frac{1}{\tau} (f_{\alpha} - f_{\alpha}^{eq}), \quad \alpha = 0, 1, \dots, 8 \quad (14)$$

where $\tilde{\mathbf{e}}_{\alpha} = (\tilde{e}_{\alpha\xi}, \tilde{e}_{\alpha\eta}) = (e_{\alpha x} \xi_x + e_{\alpha y} \xi_y, e_{\alpha x} \eta_x + e_{\alpha y} \eta_y)$ is the microscopic contravariant velocity vector in the computational plane and $\xi_x = \frac{\partial \xi}{\partial x}$, $\xi_y = \frac{\partial \xi}{\partial y}$, $\eta_x = \frac{\partial \eta}{\partial x}$ and $\eta_y = \frac{\partial \eta}{\partial y}$ are the metrics of the transformation.

3. Turbulence modeling

For the simulation of turbulent flows, two different turbulence models, namely, the Spalart–Allmaras (SA) [45] and $k - \omega$ SST [46] models are used to evaluate the turbulent eddy viscosity ν_{turb} . The SA model is governed by the following transport equation for the turbulence working variable $\hat{\nu}$:

$$\frac{\partial \hat{\nu}}{\partial t} + u_i \frac{\partial \hat{\nu}}{\partial x_i} = \frac{1}{\sigma} \left[(\nu + \hat{\nu}) \frac{\partial^2 \hat{\nu}}{\partial x_i^2} + c_{b2} \frac{\partial \hat{\nu}}{\partial x_i} \frac{\partial \hat{\nu}}{\partial x_i} \right] + c_{b1} \hat{S} \hat{\nu} - c_{\omega 1} f_{\omega} \quad (15)$$

The turbulent viscosity is then calculated as

$$\nu_{turb} = \hat{\nu} f_{v1}, \quad f_{v1} = \frac{X^3}{X^3 + c_{v1}^3}, \quad X = \frac{\hat{\nu}}{\nu} \quad (16)$$

and the other parameters are defined as

$$\begin{aligned} \hat{S} &= \Omega + \frac{\hat{\nu}}{\kappa^2 y^2} f_{v2}, \quad f_{v2} = 1 - \frac{X}{1 + X f_{v1}}, \quad g = r + c_{\omega 2} (r^5 - r) \\ r &= \min \left(\frac{\hat{\nu}}{\hat{S} \kappa^2 y^2} \right), \quad f_{\omega} = g \left[\frac{1 + c_{\omega 3}^6}{g^6 + c_{\omega 3}^6} \right]^{1/6} \\ W_{ij} &= \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right), \quad \Omega = \sqrt{2 W_{ij} W_{ij}} \end{aligned} \quad (17)$$

The closure constants used here are

$$\begin{aligned} c_{b1} &= 0.1355, \quad c_{b2} = 0.622 \\ \kappa &= 0.41, \quad \sigma = 2/3, \quad c_{v1} = 7.1 \\ c_{\omega 1} &= \frac{c_{b1}}{\kappa^2} + \frac{1 + c_{b2}}{\sigma}, \quad c_{\omega 2} = 0.3, \quad c_{\omega 3} = 2 \\ c_{t4} &= 0.5 \end{aligned} \quad (18)$$

The free-stream value of the turbulent variable is $\hat{\nu} = 3\nu_{\infty}$ and at the wall $\hat{\nu} = 0$ [47].

The transport equations for the turbulent kinetic energy k and the specific rate of dissipation ω of the $k - \omega$ SST model are given by

$$\begin{aligned} \frac{\partial k}{\partial t} + u_i \frac{\partial k}{\partial x_i} &= \left(\nu + \frac{\nu_{turb}}{\sigma_k} \right) \frac{\partial^2 k}{\partial x_i^2} + P - \beta^* \omega k \\ \frac{\partial \omega}{\partial t} + u_i \frac{\partial \omega}{\partial x_i} &= \left(\nu + \frac{\nu_{turb}}{\sigma_w} \right) \frac{\partial^2 \omega}{\partial x_i^2} + \frac{\gamma}{\nu_{turb}} P - \beta \omega^2 + 2(1 - F_1) \sigma_{w2} \frac{1}{\omega} \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j} \end{aligned} \quad (19)$$

where the production term is expressed as

$$\begin{aligned} P &= \tau_{ij} \frac{\partial u_i}{\partial x_j} \\ \tau_{ij} &= \nu_{turb} (2S_{ij}) \\ S_{ij} &= \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \end{aligned} \quad (20)$$

In [46], it is recommended that the production term in the k -equation is limited by $\min(p, 10\beta^*\omega k)$. The turbulent eddy viscosity then is computed from the following relation:

$$\nu_{turb} = \frac{a_1 k}{\max(a_1 \omega, SF_2)} \quad (21)$$

where

$$\begin{aligned} F_1 &= \tanh(\arg_1^4) \\ \arg_1 &= \min[\max(\frac{\sqrt{k}}{0.09\omega y}, \frac{500\nu}{y^2\omega}), \frac{4\sigma_{w2}k}{CD_{k\omega}y^2}] \\ CD_{k\omega} &= \max(2\sigma_{w2}\frac{1}{\omega}\frac{\partial k}{\partial x_j}\frac{\partial \omega}{\partial x_j}, 10^{-10}) \\ F_2 &= \tanh(\arg_2^2) \\ \arg_2 &= \max(\frac{2\sqrt{k}}{0.09\omega y}, \frac{500\nu}{y^2\omega}) \\ S &= \sqrt{2S_{ij}S_{ij}} \end{aligned} \quad (22)$$

The constants σ_k , σ_ω , β and γ in this model are calculated by the following relation:

$$\phi = F_1\phi_1 + (1 - F_1)\phi_2 \quad (23)$$

where constants 1 (ϕ_1) and constants 2 (ϕ_2) used here are

$$\begin{aligned} \sigma_{k1} &= 0.85, \sigma_{\omega1} = 0.5, \beta_1 = 0.075, \gamma_1 = 5/9 \\ \sigma_{k2} &= 1.0, \sigma_{\omega2} = 0.856, \beta_2 = 0.0828, \gamma_2 = 0.44 \end{aligned} \quad (24)$$

and the other constants are

$$\beta^* = 0.09, \kappa = 0.41, a_1 = 0.31 \quad (25)$$

The free-stream values of the turbulent variables are [47]

$$k_\infty = 10^{-6}u_{ref}^2, \omega_\infty = \frac{5u_{ref}}{L_{ref}} \quad (26)$$

and at the wall boundary they are as follows:

$$k_{wall} = 0, \omega_{wall} = 10\frac{6\nu}{\beta_1(\Delta d_1)^2} \quad (27)$$

where Δd_1 is the distance of the first node from the nearest wall.

To transform the given transport equations to the generalized curvilinear coordinates, the same procedure applied for the LB equation is used. The general form of the turbulent transport equations in the generalized curvilinear coordinates can be written as follows:

$$\frac{\partial M}{\partial t} + U\frac{\partial M}{\partial \xi} + V\frac{\partial M}{\partial \eta} = \left[\frac{\partial M_{v1}}{\partial \xi}\xi_x + \frac{\partial M_{v1}}{\partial \eta}\eta_x + \frac{\partial M_{v2}}{\partial \xi}\xi_y + \frac{\partial M_{v2}}{\partial \eta}\eta_y \right] + S \quad (28)$$

where $U = \xi_x u + \xi_y v$, $V = \eta_x u + \eta_y v$, M denotes the turbulence variables ($\hat{v}$, k or ω) and M_{v1} , M_{v2} and S consist of appropriate diffusive and source terms, respectively. For example, for the SA turbulence model, M_{v1} , M_{v2} and S are

$$\begin{aligned} M_{v1} &= \frac{1}{\sigma}(\nu + \hat{v})\left(\frac{\partial \hat{v}}{\partial \xi}\xi_x + \frac{\partial \hat{v}}{\partial \eta}\eta_x\right) \\ M_{v2} &= \frac{1}{\sigma}(\nu + \hat{v})\left(\frac{\partial \hat{v}}{\partial \xi}\xi_y + \frac{\partial \hat{v}}{\partial \eta}\eta_y\right) \\ S &= \frac{1}{\sigma}c_{b2}\left[\left(\frac{\partial \hat{v}}{\partial \xi}\xi_x + \frac{\partial \hat{v}}{\partial \eta}\eta_x\right)^2 + \left(\frac{\partial \hat{v}}{\partial \xi}\xi_y + \frac{\partial \hat{v}}{\partial \eta}\eta_y\right)^2\right] + c_{b1}\hat{S}\hat{v} - c_{\omega1}f_\omega \end{aligned} \quad (29)$$

The coupling of the turbulence models and the LB equation is rather straightforward. After computing the turbulent eddy viscosity from the turbulence models, the effective viscosity is calculated by

$$\nu_{eff} = \nu + \nu_{turb} \quad (30)$$

Hence, the effect of turbulence is introduced into the LB equation by considering the relaxation time (Eq. (13)) calculated based on the effective viscosity as follows:

$$\tau = \frac{\nu_{eff}}{\gamma c_s^2} \quad (31)$$

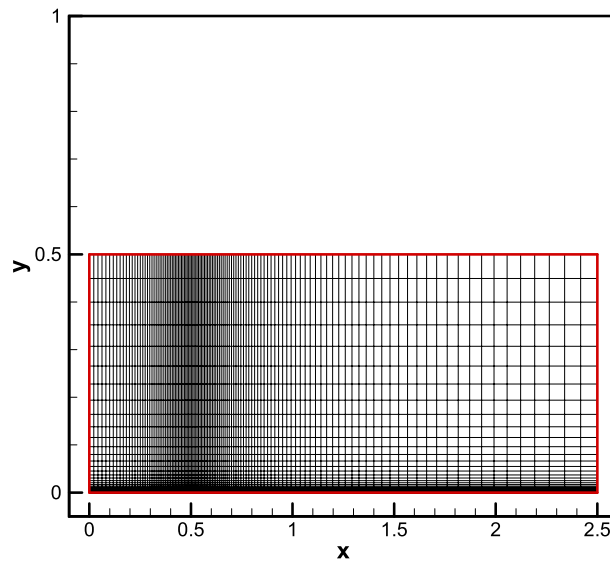


Fig. 1. A (121×61) mesh distribution for turbulent flow over flat plate.

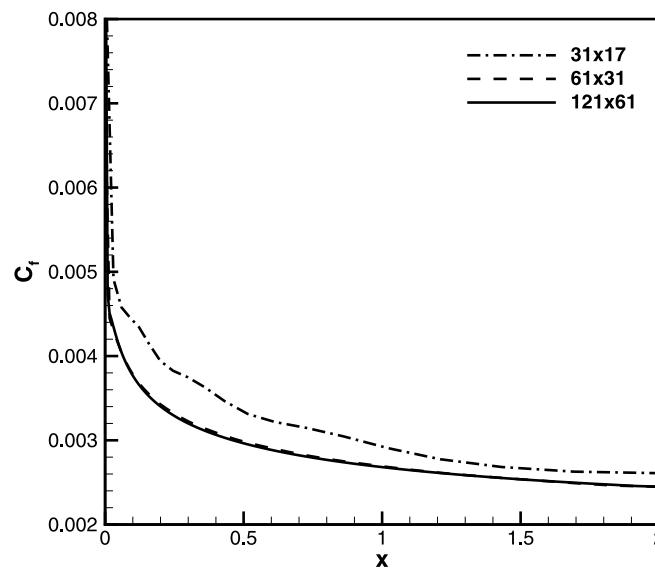


Fig. 2. Grid refinement study on skin friction coefficient distribution for turbulent flow over flat plate at $Re = 1.0 \times 10^7$ with SA turbulence model.

4. Discretization procedure

In the present study, the fifth-order finite-difference weighted essentially non-oscillatory (WENO) scheme is applied to discretize the spatial derivatives in the preconditioned LB equation in the computational domain and the first-order implicit–explicit Runge–Kutta scheme is used for the temporal discretization to achieve an accurate and efficient incompressible turbulent flow solver. For turbulence models, the convective terms of the transport equations in the computational plane are discretized by the third-order WENO scheme and the second-order central difference scheme is used for the diffusive terms. For the temporal discretization, a simple first-order explicit scheme is used.

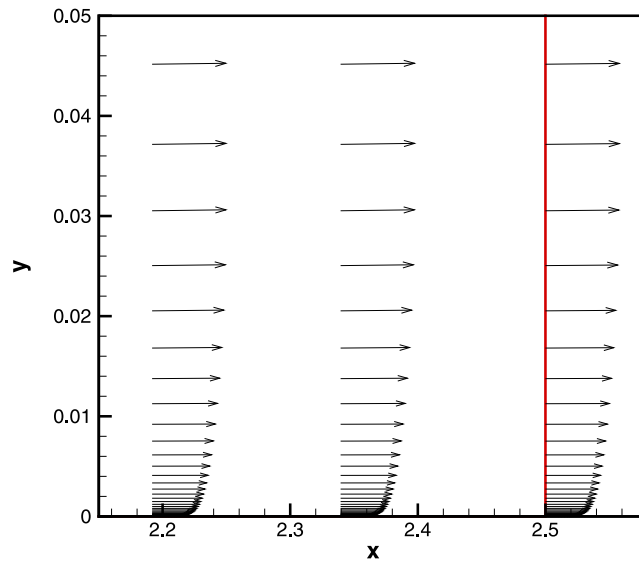


Fig. 3. Computed flow field shown by velocity profiles for turbulent flow over flat plate at $Re = 1.0 \times 10^7$ with SA turbulence model.

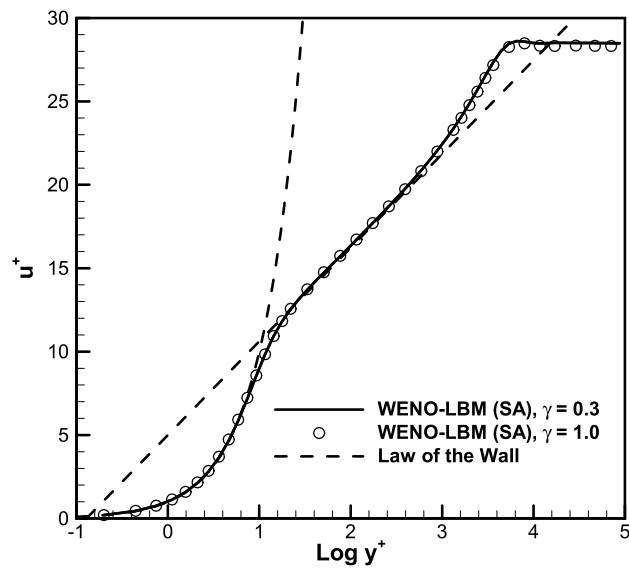


Fig. 4. Comparison of velocity profile for turbulent flow over flat plate at $Re = 1.0 \times 10^7$ with SA turbulence model.

Table 1

Order of accuracy of solution based on L_2 -norm of C_f compared with most refined grid (121×61) for turbulent flow over flat plate at $Re = 1.0 \times 10^7$ with SA turbulence model.

Grid size	Log (L_2 -norm)
(31×17)	−4.347
(61×31)	−5.819
Order of accuracy	~4.88

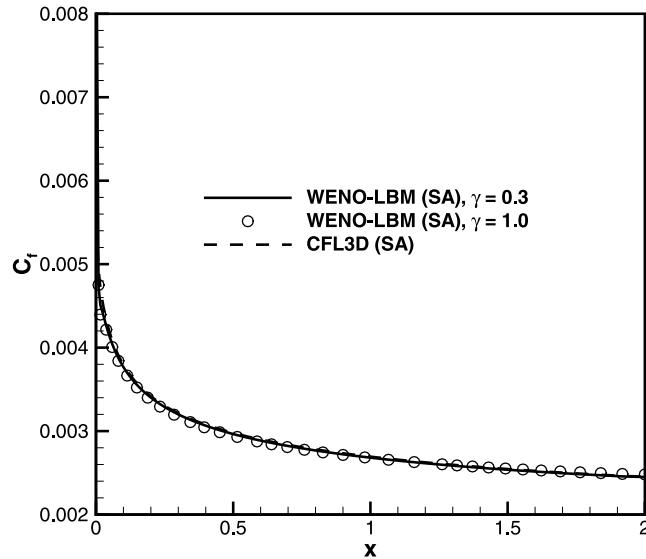


Fig. 5. Comparison of skin friction coefficient distribution for turbulent flow over flat plate at $Re = 1.0 \times 10^7$ with SA turbulence model.

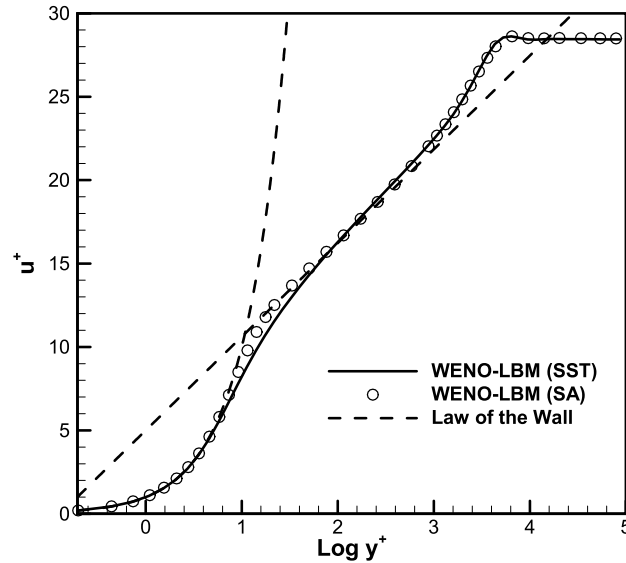


Fig. 6. Comparison of velocity profile for turbulent flow over flat plate at $Re = 1.0 \times 10^7$.

4.1. Spatial discretization

4.1.1. Fifth-order WENO scheme

The natural implementation of upwinding in the WENO scheme, is to use the signs of the microscopic contravariant velocity in front of the spatial derivatives in Eq. (14). The semi-discretized form of Eq. (14) is

$$\frac{\partial f_{\alpha,i,j}}{\partial t} + \frac{\tilde{e}_{\alpha\xi}}{\Delta\xi}(f_{\alpha,i+\frac{1}{2},j} - f_{\alpha,i-\frac{1}{2},j}) + \frac{\tilde{e}_{\alpha\eta}}{\Delta\eta}(f_{\alpha,i,j+\frac{1}{2}} - f_{\alpha,i,j-\frac{1}{2}}) = -\frac{1}{\tau}(f_{\alpha,i,j} - f_{\alpha,i,j}^{eq}) \quad (32)$$

where i and j are the indices of each cell and $f_{\alpha,i\pm\frac{1}{2},j}$ and $f_{\alpha,i,j\pm\frac{1}{2}}$ are calculated based on the fifth-order WENO scheme [48]. For example, $f_{\alpha,i+\frac{1}{2},j}$ is calculated as follows:

$$f_{\alpha,i+\frac{1}{2},j} = \sum_{k=1}^3 \omega_k f_{\alpha,i+\frac{1}{2},j}^{(k)} \quad (33)$$

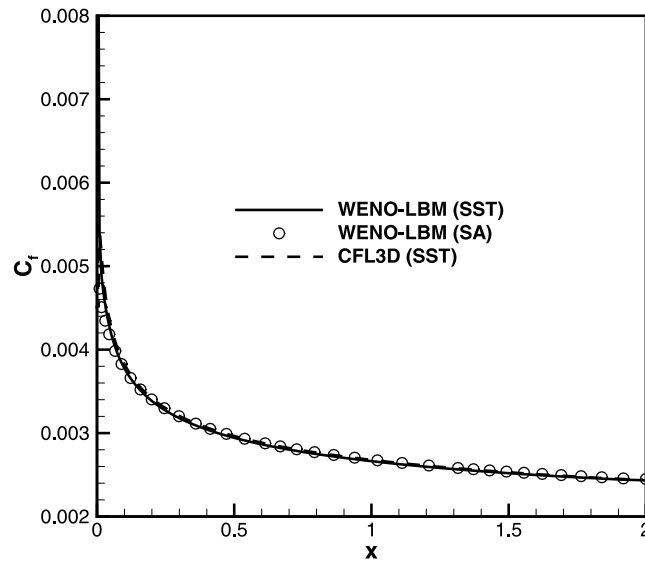


Fig. 7. Comparison of skin friction coefficient distribution for turbulent flow over flat plate at $Re = 1.0 \times 10^7$.

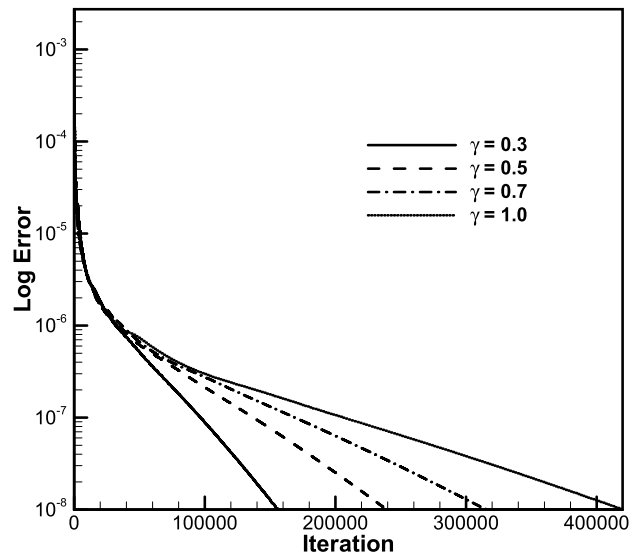


Fig. 8. Effect of preconditioning on convergence rate of solution for turbulent flow over flat plate at $Re = 1.0 \times 10^7$ with SA turbulence model.

where $f_{\alpha, i+\frac{1}{2}, j}^{(k)}$ is a third-order approximation of f_{α} (and if $\tilde{e}_{\alpha\xi} > 0$), we have

$$\begin{aligned} f_{\alpha, i+\frac{1}{2}, j}^{(1)} &= \frac{1}{3}f_{\alpha, i-2, j} - \frac{7}{6}f_{\alpha, i-1, j} + \frac{11}{6}f_{\alpha, i, j} \\ f_{\alpha, i+\frac{1}{2}, j}^{(2)} &= -\frac{1}{6}f_{\alpha, i-1, j} + \frac{5}{6}f_{\alpha, i, j} + \frac{1}{3}f_{\alpha, i+1, j} \\ f_{\alpha, i+\frac{1}{2}, j}^{(3)} &= \frac{1}{3}f_{\alpha, i, j} + \frac{5}{6}f_{\alpha, i+1, j} - \frac{1}{6}f_{\alpha, i+2, j} \end{aligned} \quad (34)$$

The nonlinear weights ω_k in Eq. (33) are

$$\omega_l = \omega_l / \sum_{l=1}^3 \omega_l, \quad \omega_l = \frac{\gamma_l}{(\varepsilon + \beta_l)^2} \quad (35)$$

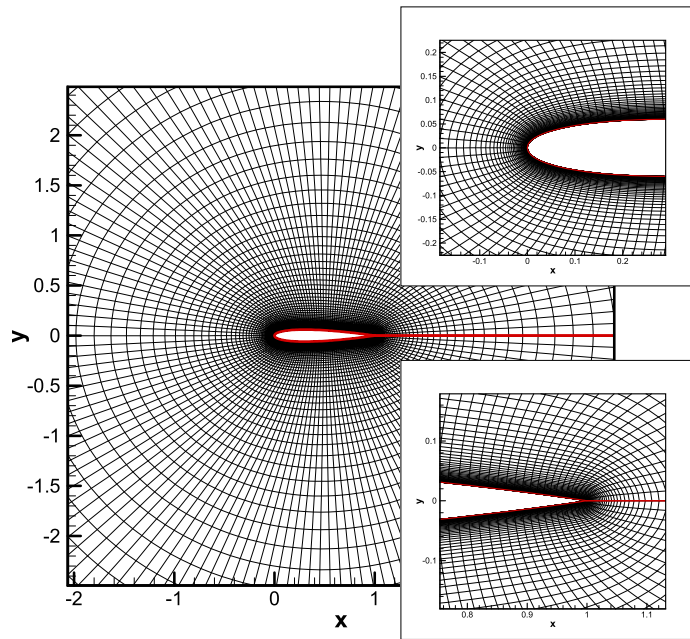


Fig. 9. A (151 × 121) mesh distribution for turbulent flow over NACA0012 airfoil.

where the smoothness indicators β_l ($l = 1, 2, 3$) are given by

$$\begin{aligned}\beta_1 &= \frac{13}{12}(f_{\alpha,i-2} - 2f_{\alpha,i-1,j} + f_{\alpha,i,j})^2 + \frac{1}{4}(f_{\alpha,i-2,j} - 4f_{\alpha,i-1,j} + 3f_{\alpha,i,j})^2 \\ \beta_2 &= \frac{13}{12}(f_{\alpha,i-1,j} - 2f_{\alpha,i,j} + f_{\alpha,i+1,j})^2 + \frac{1}{4}(f_{\alpha,i-1,j} - f_{\alpha,i+1,j})^2 \\ \beta_3 &= \frac{13}{12}(f_{\alpha,i,j} - 2f_{\alpha,i+1,j} + f_{\alpha,i+2,j})^2 + \frac{1}{4}(3f_{\alpha,i,j} - 4f_{\alpha,i+1,j} + f_{\alpha,i+2,j})^2\end{aligned}\quad (36)$$

and the linear weights γ_l are $\gamma_1 = 0.1$, $\gamma_2 = 0.6$ and $\gamma_3 = 0.3$. Also, ε is a small positive number used to avoid the denominator becoming zero (here, $\varepsilon = 10^{-6}$).

Eq. (33) is applied in the computational domain for the grid points $i = 3, \dots, I_{max} - 2$ for each j . At the near boundary points, the third-order reconstructions for $f_{\alpha,i+\frac{1}{2},j}$ are used as follows:

$$\begin{aligned}\text{for } i = 2: \quad f_{\alpha,i+\frac{1}{2},j} &= -\frac{1}{6}f_{\alpha,i-1,j} + \frac{5}{6}f_{\alpha,i,j} + \frac{1}{3}f_{\alpha,i+1,j} \\ \text{for } i = I_{max} - 1: \quad f_{\alpha,i+\frac{1}{2},j} &= -\frac{1}{6}f_{\alpha,i-1,j} + \frac{5}{6}f_{\alpha,i,j} + \frac{1}{3}f_{\alpha,i+1,j}\end{aligned}\quad (37)$$

and if $\tilde{e}_{\alpha\xi} < 0$, $f_{\alpha,i+\frac{1}{2},j}$ is constructed symmetrically as above at $i + \frac{1}{2}, j$. The calculation of $f_{\alpha,i,j\pm\frac{1}{2}}$ is performed in the same manner.

4.1.2. Third-order WENO scheme

For the third-order WENO scheme, the derivative $\frac{\partial M}{\partial \xi}$ in Eq. (28) is calculated as follows:

$$\frac{\partial M}{\partial \xi} = \frac{(M_{i+\frac{1}{2},j} - M_{i-\frac{1}{2},j})}{\Delta \xi}\quad (38)$$

If $U > 0$ then $M_{i+\frac{1}{2},j}$ is defined by

$$M_{i+\frac{1}{2},j} = \sum_{k=1}^2 \omega_k M_{i+\frac{1}{2},j}^{(k)}\quad (39)$$

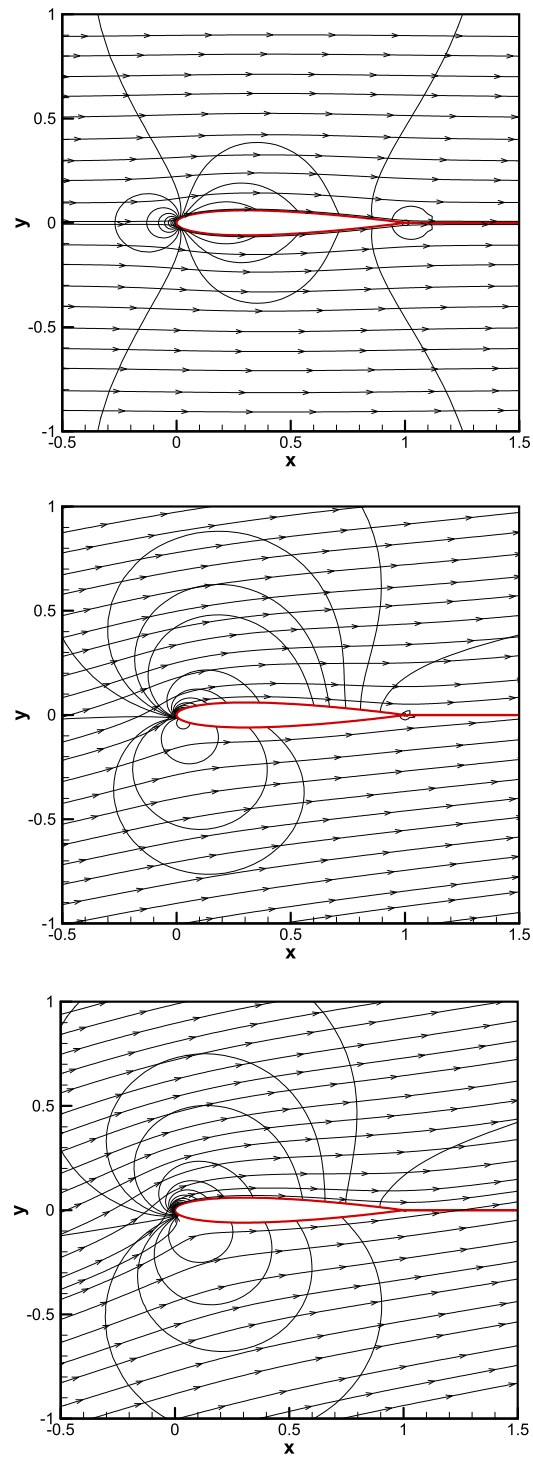


Fig. 10. Computed flow field shown by pressure contours and streamlines for turbulent flow over NACA0012 airfoil at $Re = 6.0 \times 10^6$ and $\alpha = 0^\circ$ (top), $\alpha = 10^\circ$ (middle) and $\alpha = 15^\circ$ (bottom) with SA turbulence model.

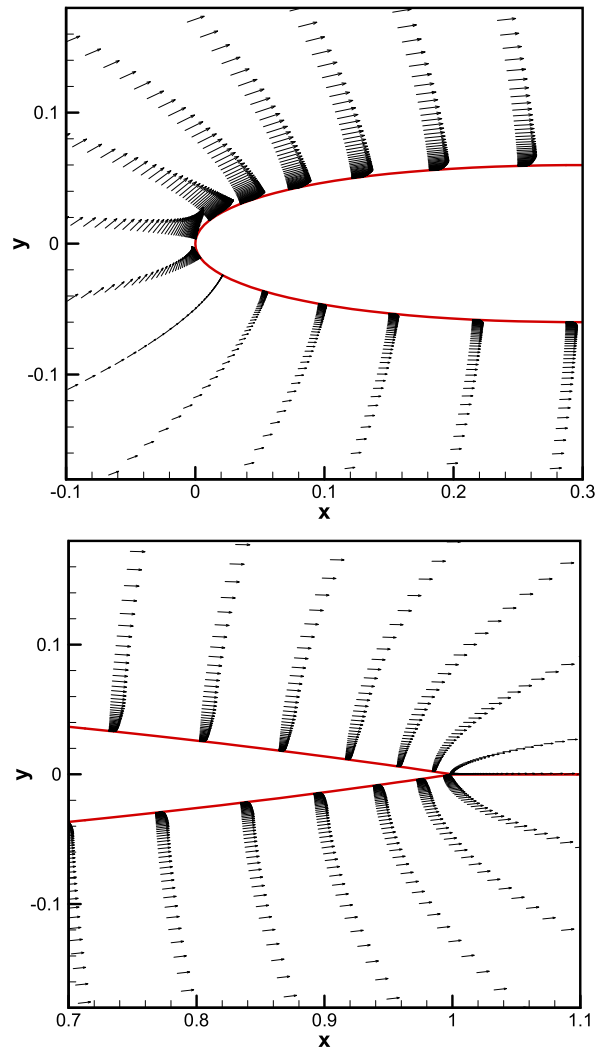


Fig. 11. Computed flow field shown by velocity profiles near the leading (top) and trailing (bottom) edges for turbulent flow over NACA0012 airfoil at $Re = 6.0 \times 10^6$ and $\alpha = 10^\circ$ with SA turbulence model.

where $M_{i+\frac{1}{2}}^{(k)}$ is a second approximation of M , we have

$$\begin{aligned} M_{i+\frac{1}{2}}^{(1)} &= -\frac{1}{2}M_{i-1,j} + \frac{3}{2}M_{i,j} \\ M_{i+\frac{1}{2}}^{(2)} &= \frac{1}{2}M_{i,j} + \frac{1}{2}M_{i+1,j} \end{aligned} \quad (40)$$

The nonlinear weights ω_k in Eq. (39) are

$$\omega_l = \omega_l / \sum_{l=1}^2 \omega_l, \quad \omega_l = \frac{\gamma_l}{(\varepsilon + \beta_l)^2} \quad (41)$$

where the smoothness indicators β_l ($l = 1, 2$) are given by

$$\begin{aligned} \beta_1 &= (M_{i,j} - M_{i-1,j})^2 \\ \beta_2 &= (M_{i+1,j} - M_{i,j})^2 \end{aligned} \quad (42)$$

and the linear weights γ_l are $\gamma_1 = 1/3$ and $\gamma_2 = 2/3$.

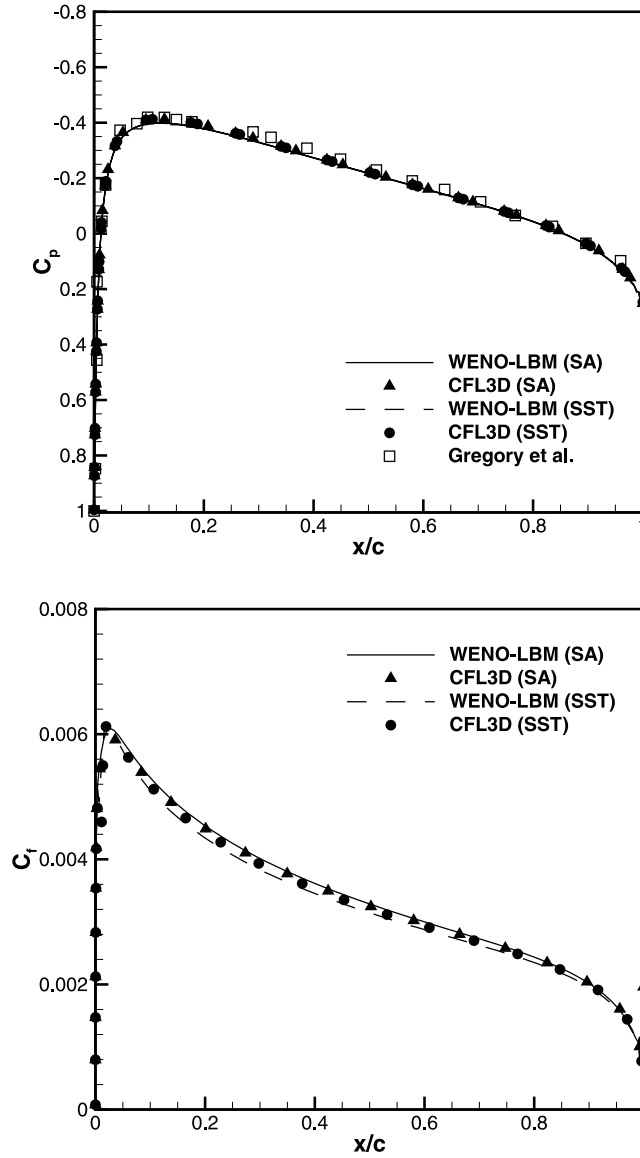


Fig. 12. Comparison of surface pressure coefficient distribution (top) and skin friction coefficient distribution (bottom) for turbulent flow over NACA0012 airfoil at $Re = 6.0 \times 10^6$ and $\alpha = 0^\circ$.

4.2. Temporal discretization

For the temporal discretization, the first-order implicit-explicit (IMEX) Runge-Kutta scheme as described in [2,5] is applied to the LB equation to remove the restriction on Δt . Also, a simple first-order explicit scheme is used for the time discretization of the turbulence transport equations. Since the steady-state solution is of concern, the order of accuracy of the time discretization scheme is not an important issue here.

5. Boundary conditions implementation

The implementation of the boundary conditions in the WENO-LBM for the calculation of the distribution function f_α on the boundaries in the present study is different with that given in Ref. [5]. Here, the distribution function f_α at the boundary nodes is calculated directly based on the solution of Eq. (14) with the note that the fifth-order one-sided finite difference formulation is used to calculate the derivatives $\frac{\partial f_\alpha}{\partial \xi}$ and $\frac{\partial f_\alpha}{\partial \eta}$. For example, the calculation of $\frac{\partial f_\alpha}{\partial \xi}$ at the boundary nodes $i = 1$

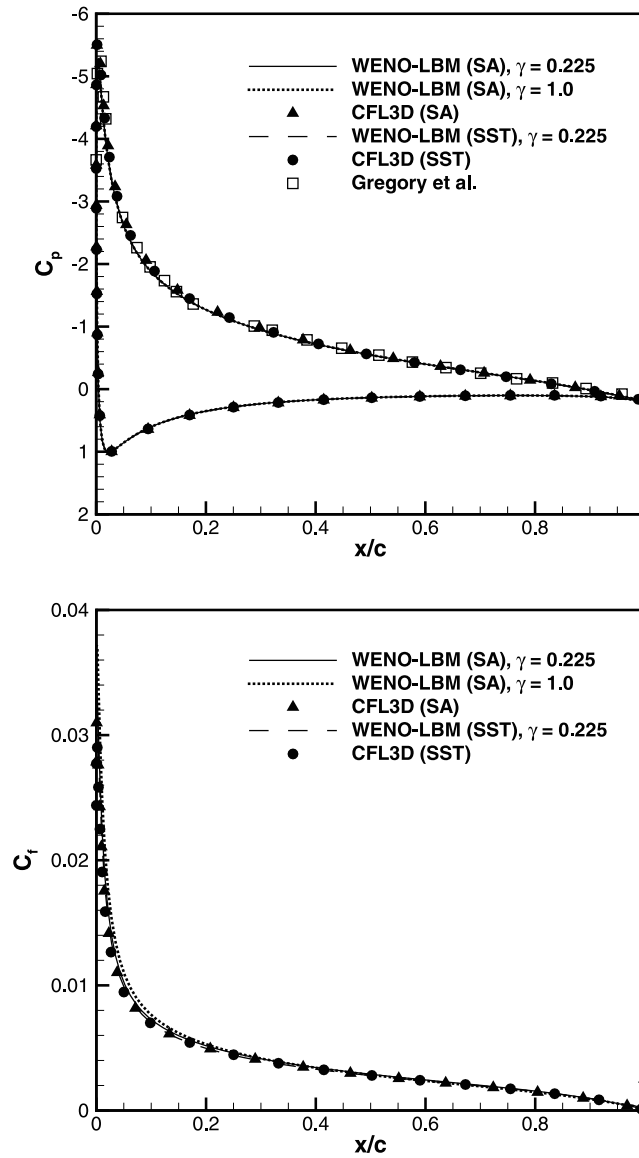


Fig. 13. Comparison of surface pressure coefficient distribution (top) and skin friction coefficient distribution on the upper surface (bottom) for turbulent flow over NACA0012 airfoil at $Re = 6.0 \times 10^6$ and $\alpha = 10^\circ$.

and $I_{\max}$ is performed as follows:

$$\begin{aligned} \frac{\partial f_{\alpha,1}}{\partial \xi} &= \left(\frac{1}{60\Delta\xi} \right) (-137f_{\alpha,1} + 300f_{\alpha,2} - 300f_{\alpha,3} + 200f_{\alpha,4} - 75f_{\alpha,5} + 12f_{\alpha,6}) \\ \frac{\partial f_{\alpha,I_{\max}}}{\partial \xi} &= \left(\frac{-1}{60\Delta\xi} \right) (-137f_{\alpha,I_{\max}} + 300f_{\alpha,I_{\max}-1} - 300f_{\alpha,I_{\max}-2} + 200f_{\alpha,I_{\max}-3} - 75f_{\alpha,I_{\max}-4} + 12f_{\alpha,I_{\max}-5}) \end{aligned} \quad (43)$$

The equilibrium distribution function f_{α}^{eq} required for calculating the distribution function f_{α} in Eq. (14) is known at the boundaries and it is calculated by Eq. (5) with the use of the specified/extrapolated macroscopic variables at the desired boundary. At the inlet and on the wall boundary with no-slip condition, the velocity components are known and the one-sided finite difference extrapolation is used to discretize $\frac{\partial p}{\partial \eta} = 0$ to calculate the pressure at the boundary as follows:

$$p_1 = \frac{1}{125} (240p_2 - 180p_3 + 80p_4 - 15p_5) \quad (44)$$

At the outlet, the pressure is set to a fixed value and the same extrapolation is used for the velocity components.

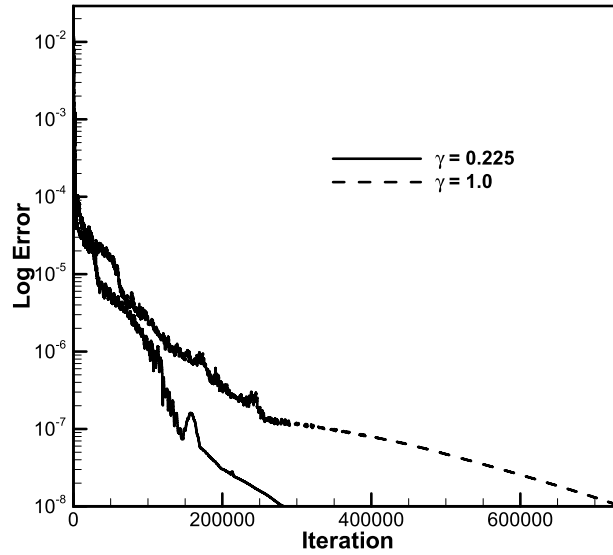


Fig. 14. Effect of preconditioning on convergence rate of solution for turbulent flow over NACA0012 airfoil at $Re = 6.0 \times 10^6$ and $\alpha = 10^\circ$ with SA turbulence model.

6. Numerical results

To examine the accuracy and robustness of the preconditioned WENO-LBM, the solution algorithm is used to simulate two incompressible turbulent flow problems: The first test case is the 2D turbulent flow over a flat plate at $Re = 1.0 \times 10^7$ and the second one is the 2D turbulent flow over a NACA0012 airfoil at $Re = 6.0 \times 10^6$ and different angles of attack. The computations are carried out on a single core of an i7-5500 (2.40 GHz) CPU and the convergence of the steady-state solution is obtained by considering the L_∞ -norm of the u -velocity profile set to be 10^{-8} . Results obtained for these test cases are compared with the available numerical and experimental results.

6.1. Turbulent flow over a flat plate

The 2D incompressible turbulent flow over a flat plate is considered first. Herein, the numerical simulation is carried out for the Reynolds number $Re = \frac{u_{ref} L}{\nu} = 1.0 \times 10^7$ and the Mach number $Ma = \frac{u_{ref}}{c_s} = 0.2$, where $L = 2$ is the plate length. The numerical results of the CFL3D code [49] with the same Re and Ma numbers are used for comparisons of the present results. The CFL3D code is a structured-grid, cell-centered, upwind-biased, Reynolds-averaged Navier–Stokes (RANS) code developed at NASA Langley Research Center. A 121×61 non-uniform mesh used here is shown in Fig. 1. The flat plate starts at $x = 0.5$ and the symmetry boundary condition is applied for $0 \leq x < 0.5$. The grid lines are clustered near the wall surface to accurately resolve the turbulent boundary layer. According to [20], based on the linear stability analysis, the optimal preconditioning parameter γ lies in the range of $(1.0Ma, 1.5Ma)$. Thus, for this test case, the preconditioning parameter is set to $\gamma = 1.5Ma = 0.3$ and below this value the solution may become unstable. Also, the local time-stepping method $\Delta t_{i,j} = \min(0.2 \times \min(\Delta x_{i,j}, \Delta y_{i,j}), 0.0004)$ is used to further accelerate the solution to the steady-state condition.

A grid independence study is performed for the flat plate case with the SA turbulence model. Three different grid sizes, namely, (31×17) , (61×31) and (121×61) are used and the distribution of the skin friction coefficient for the flat plate for these three grids is shown in Fig. 2. It is observed that the grids (61×31) and (121×61) provide similar results and they are appropriate for an accurate calculation of the turbulent boundary layer over the flat plate at $Re = 1.0 \times 10^7$. Table 1 gives the calculated error based on the L_2 -norm of the skin friction coefficient on the flat plate compared with the solution of the most refined grid (121×61) and the fifth-order accuracy of the solution of the WENO-LBM is verified.

Fig. 3 gives the computed flow field shown by the velocity profiles predicted by the SA turbulence model. The near wall behavior of the flow field in terms of the non-dimensional velocity profile $u^+ = u/u_\tau$ vs. $y^+ = yu_\tau/\nu$ is shown in Fig. 4 for the SA turbulence model with the preconditioning parameter $\gamma = 0.3$ and without the preconditioning ($\gamma = 1.0$). It is observed that the results obtained by the WENO-LBM is in excellent agreement with the law of the wall [50] and the results of the preconditioned and non-preconditioned LB equations are almost the same. Fig. 5 shows the comparison of the skin friction coefficient distribution over the flat plate using the SA turbulence model with the numerical results of the CFL3D code [49] which indicates good agreement. Figs. 6 and 7 show the results using the $k-\omega$ SST turbulence model and again the present results are in good agreement with the law of the wall and the numerical results of the CFL3D code, with the note that the CFL3D results are based on the (545×385) grid size which is very finer than that used in the WENO-LBM. Although

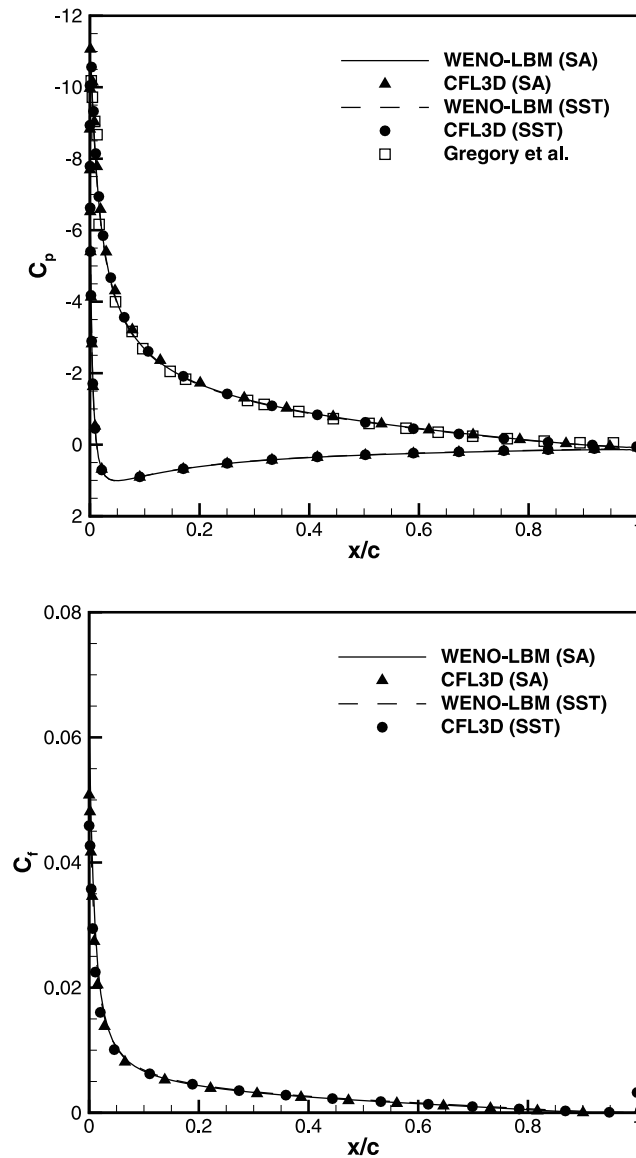


Fig. 15. Comparison of surface pressure coefficient distribution (top) and skin friction coefficient distribution on the upper surface (bottom) for turbulent flow over NACA0012 airfoil at $Re = 6.0 \times 10^6$ and $\alpha = 15^\circ$.

not shown here, the present results are also in good agreement with those of the second-order finite volume LBM [41]. Note that the number of grid points used in Ref. [41] is the same as that used in the CFL3D code which demonstrates the accuracy of the preconditioned WENO-LBM applied.

A sensitivity study is performed to examine the effect of the preconditioning parameter on the convergence rate of the solution. Fig. 8 shows the convergence rate of the solution of the WENO-LBM with the SA turbulence model based on the L_∞ -norm of the u -velocity profile using different values of the preconditioning parameter, $\gamma = 0.3, 0.5, 0.7$ and $\gamma = 1.0$ (no preconditioning). It is observed that by using the preconditioning technique the convergence rate of the solution is significantly increased and $\gamma = 0.3$ gives a better performance.

Table 2 gives the computation time of the steady-state solution of the preconditioned WENO-LBM for the turbulent flow over the flat plate at $Re = 1.0 \times 10^7$ with the SA turbulence model for the three grids used with the preconditioning parameter $\gamma = 0.3$. It is indicated that the incompressible turbulent flow solver based on the WENO-LBM converges with reasonable CPU times for these three grids.

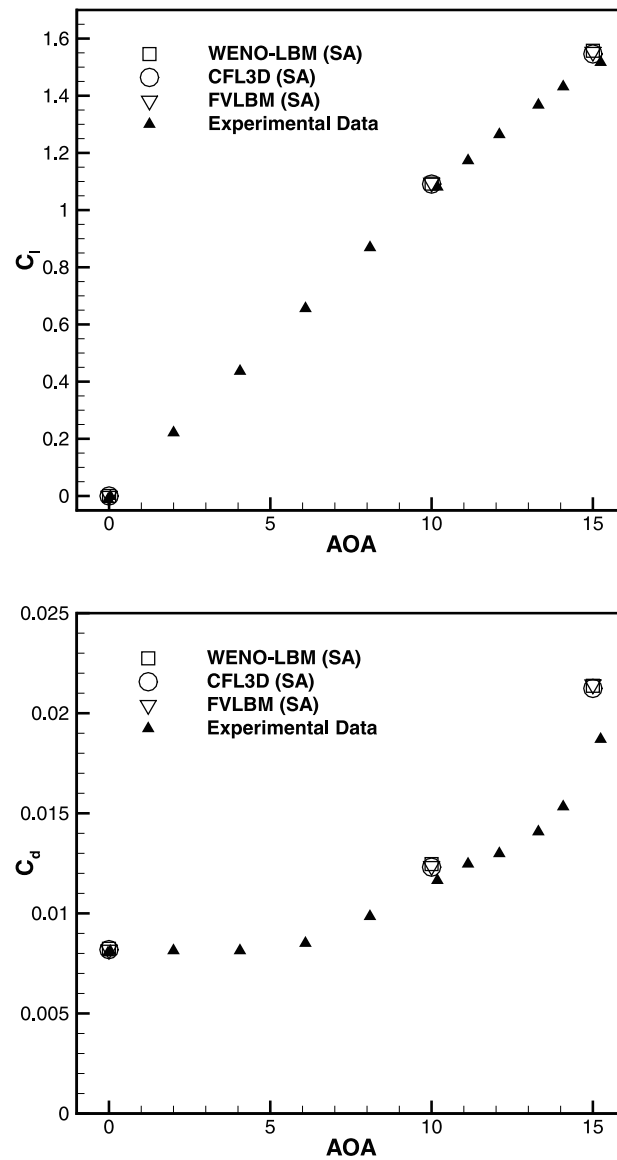


Fig. 16. Comparison of lift (top) and drag (bottom) coefficients for turbulent flow over NACA0012 airfoil at $Re = 6.0 \times 10^6$ with SA turbulence model.

Table 2

Computation time of solution of turbulent flow over flat plate at $Re = 1.0 \times 10^7$ with SA turbulence model.

Grid size	CPU time (s)
(31×17)	29.4
(61×31)	165.4
(121×61)	1105.8

6.2. Turbulent flow over a NACA0012 airfoil

For the second test case, the 2D incompressible turbulent flow over the NACA0012 airfoil is considered to demonstrate the accuracy and robustness of the preconditioned WENO-LBM in the generalized curvilinear coordinates. In this case, an O-type grid with the size of 151×121 is used and the grid is clustered around the airfoil surface to accurately resolve the turbulent boundary layer near the airfoil surface. Fig. 9 shows the mesh used for this simulation. The Reynolds number $Re = \frac{u_{ref} c}{\nu}$ is defined based on the free stream velocity u_{ref} and the chord length of the airfoil $c = 1$ and the free stream

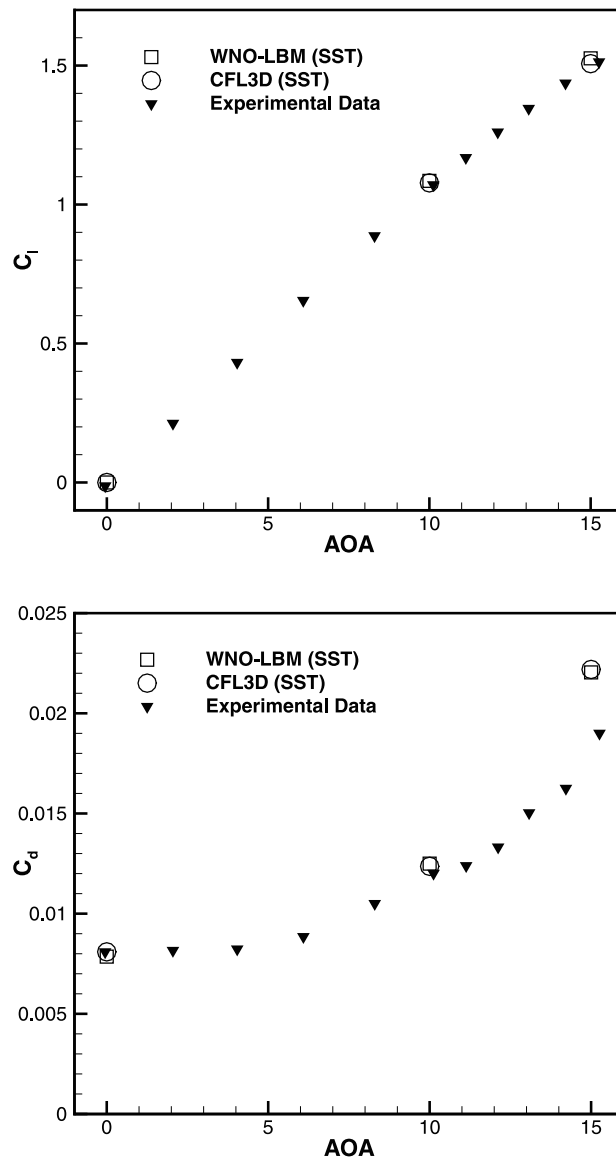


Fig. 17. Comparison of lift (top) and drag (bottom) coefficients for turbulent flow over NACA0012 airfoil at $Re = 6.0 \times 10^6$ with $k - \omega$ SST turbulence model.

Mach number is $Ma = \frac{u_{ref}}{c_s}$. Here, the steady-state solutions are obtained at $Re = 6.0 \times 10^6$, $Ma = 0.15$ and three angles of attack $\alpha = 0^\circ, 10^\circ, 15^\circ$. The local time step is used as $\Delta t_{i,j} = \min(0.2 \times \min(\Delta x_{i,j}, \Delta y_{i,j}), 0.0004)$ and the preconditioning parameter is set to be $\gamma = 1.5Ma = 0.225$ for $\alpha = 0^\circ, 10^\circ$ and $\gamma = 2Ma = 0.3$ for $\alpha = 15^\circ$. Note that below these values the solution may become unstable.

The pressure contours and the streamlines obtained by the present solution for the flow around the NACA0012 airfoil at the angles of attack $\alpha = 0^\circ, 10^\circ$ and 15° with the SA and $k - \omega$ SST turbulence models are almost identical, and therefore, only the results of the SA turbulence model are shown in Fig. 10. Fig. 11 gives the computed flow field shown by the velocity profiles near the leading and trailing edges of the airfoil at $\alpha = 10^\circ$ and it is observed that the flow is attached for this condition. Fig. 12 compares the distribution of the surface pressure and skin friction coefficients on the upper surface of the airfoil at $\alpha = 0^\circ$ using the two turbulence models with the experimental measurements of Gregory et al. [51] and also the numerical results of the CFL3D code [49] and Fig. 13 compares the results at $\alpha = 10^\circ$. It is observed that the results obtained by the WENO-LBM agree well with those of Refs. [49,51] and the accuracy of the present solution is not sensitive to the value of the preconditioning parameter. A sensitivity study is now performed to examine the effect of the preconditioning parameter on the convergence rate of the solution. Fig. 14 shows the convergence rate of the solution of the WENO-LBM for the turbulent flow over the NACA0012 airfoil at $\alpha = 10^\circ$ with the SA turbulence model. The error is calculated based

on the L_∞ -norm of the u -velocity profile for the preconditioning parameter $\gamma = 0.225$ and no preconditioning $\gamma = 1.0$. It is obvious that the preconditioning considerably improves the convergence rate of the solution. Fig. 15 gives the present results at the angle of attack $\alpha = 15^\circ$ and good agreement is obtained compared to those of Refs. [49,51]. The lift and drag coefficients obtained by the preconditioned WENO-LBM using the SA and $k - \omega$ SST turbulence models are compared with those of the CFL3D code [49], the second-order finite volume LBM [41] and also the experiment of Ladson [52], as shown in Figs. 16 and 17. It is observed that the results obtained by the WENO-LBM are comparable with the experimental data and the mentioned numerical results, with the note that the numerical results of the CFL3D code and the finite volume LBM are obtained on a very fine grid with the size 897×257 and 513×257 , respectively, which are more refined ones than that used in the present solution method. The CPU time of the steady-state solution of the preconditioned WENO-LBM for the turbulent flow over the NACA0012 airfoil at $Re = 6.0 \times 10^6$ and $\alpha = 10^\circ$ with the SA turbulence model is about 4953 s.

7. Conclusions

In this study, a preconditioned weighted essentially non-oscillatory finite-difference LBM (WENO-LBM) solver for computing 2D incompressible turbulent flows in the generalized curvilinear coordinate is applied and assessed. The spatial derivatives of the preconditioned LB equation are discretized using the fifth-order WENO scheme and an implicit–explicit Runge–Kutta scheme is adopted for the time discretization. Two turbulence models, namely, the Spalart–Allmaras (SA) and $k - \omega$ SST models are used. Two test cases namely, the 2D turbulent flow over a flat plate and the 2D turbulent flow over a NACA0012 airfoil are simulated to investigate the accuracy and robustness of the high-order accurate preconditioned WENO-LBM solver developed. Some conclusions and remarks regarding the present study are as follows:

- 1- It is shown that the results obtained by the present solution method are in good agreement with the available numerical and experimental results for the problems simulated. The study shows that the preconditioned WENO-LBM provides accurate solutions and the grid sizes used in the present solution method are much coarser than those used in the Navier–Stokes and LB solvers.
- 2- A sensitivity study performed to examine the effect of the preconditioning parameter γ on the accuracy and convergence rate of the solution. It is observed that the preconditioning can significantly improve the convergence rate of the solution and the steady-state solution is not sensitive to the value of preconditioning parameter.
- 3- It is indicated that the two turbulence models applied in the preconditioned WENO-LBM provide nearly the same results for the two problems simulated. The SA turbulence model has a simpler formulation than the $k - \omega$ SST turbulence model and thus it is preferable to use this turbulence model for such flow problems. The accuracy and performance of these two turbulence models applied in the preconditioned WENO-LBM can be assessed for other flow problems at different conditions.
- 4- The study shows that the present solution procedure is robust, efficient and accurate for solving steady incompressible turbulent flows. Results obtained by the present solution are comparable with those of the Navier–Stokes solvers with the note that the implementation of the WENO finite-difference LBM is much simpler, and thus, it can be considered as a suitable alternative turbulent flow solver to high-order finite difference Reynolds-averaged Navier–Stokes solvers for accurately simulating turbulent flows over practical problems.

Acknowledgment

The authors would like to thank Sharif University of Technology for the support of this research.

References

- [1] X. He, L.S. Luo, Theory of the lattice Boltzmann method: From the Boltzmann equation to the lattice Boltzmann equation, *Phys. Rev. E* 56 (6) (1997) 6811.
- [2] Y. Wang, Y.L. He, T.S. Zhao, G.H. Tang, W.Q. Tao, Implicit-explicit finite-difference lattice Boltzmann method for compressible flows, *Internat. J. Modern Phys. C* 18 (12) (2007) 1961–1983.
- [3] R. Mei, W. Shyy, On the finite difference-based lattice Boltzmann method in curvilinear coordinates, *J. Comput. Phys.* 143 (2) (1998) 426–448.
- [4] Z. Guo, T.S. Zhao, Explicit finite-difference lattice Boltzmann method for curvilinear coordinates, *Phys. Rev. E* 67 (6) (2003) 066709.
- [5] K. Hejranfar, M.H. Saadat, S. Taheri, High-order weighted essentially nonoscillatory finite-difference formulation of the lattice Boltzmann method in generalized curvilinear coordinates, *Phys. Rev. E* 95 (2) (2017) 023314.
- [6] K. Hejranfar, E. Ezzatneshan, A high-order compact finite-difference lattice Boltzmann method for simulation of steady and unsteady incompressible flows, *Internat. J. Numer. Methods Fluids* 75 (10) (2014) 713–746.
- [7] K. Hejranfar, E. Ezzatneshan, Implementation of a high-order compact finite-difference lattice Boltzmann method in generalized curvilinear coordinates, *J. Comput. Phys.* 267 (2014) 28–49.
- [8] M. Stiebler, J. Tölke, M. Krafczyk, An upwind discretization scheme for the finite volume lattice Boltzmann method, *Comput. & Fluids* 35 (8) (2006) 814–819.
- [9] D.V. Patil, K.N. Lakshmisha, Finite volume TVD formulation of lattice Boltzmann simulation on unstructured mesh, *J. Comput. Phys.* 228 (14) (2009) 5262–5279.
- [10] T. Lee, C.L. Lin, A characteristic Galerkin method for discrete Boltzmann equation, *J. Comput. Phys.* 171 (1) (2001) 336–356.
- [11] Y. Li, E.J. LeBoeuf, P.K. Basu, Least-squares finite-element scheme for the lattice Boltzmann method on an unstructured mesh, *Phys. Rev. E* 72 (4) (2005) 046711.

- [12] A. Düster, L. Demkowicz, E. Rank, High-order finite elements applied to the discrete Boltzmann equation, *Internat. J. Numer. Methods Engrg.* 67 (8) (2006) 1094–1121.
- [13] X. Shi, J. Lin, Z. Yu, Discontinuous Galerkin spectral element lattice Boltzmann method on triangular element, *Internat. J. Numer. Methods Fluids* 42 (11) (2003) 1249–1261.
- [14] M. Min, T. Lee, A spectral-element discontinuous Galerkin lattice Boltzmann method for nearly incompressible flows, *J. Comput. Phys.* 230 (1) (2011) 245–259.
- [15] K. Hejranfar, M. Hajihassanpour, Chebyshev collocation spectral lattice Boltzmann method for simulation of low-speed flows, *Phys. Rev. E* 91 (2015) 013301.
- [16] K. Hejranfar, M. Hajihassanpour, Chebyshev collocation spectral lattice Boltzmann method in generalized curvilinear coordinates, *Comput. & Fluids* 146 (2017) 154–173.
- [17] Hejranfar, A. Ghaffarian, A spectral difference lattice Boltzmann method for solution of inviscid compressible flows on structured grids, *Comput. Math. Appl.* 72 (2016) 1341–1368.
- [18] P.R. Rao, L.A. Schaefer, Numerical stability of explicit off-lattice Boltzmann schemes: A comparative study, *J. Comput. Phys.* 285 (2015) 251–264.
- [19] Z. Guo, T.S. Zhao, Y. Shi, Preconditioned lattice-Boltzmann method for steady flows, *Phys. Rev. E* 70 (6) (2004) 066706.
- [20] S. Izquierdo, N. Fuego, Optimal preconditioning of lattice Boltzmann methods, *J. Comput. Phys.* 228 (17) (2009) 6479–6495.
- [21] A. Ten Cate, E. Van Vliet, J.J. Derksen, H.E.A. Van den Akker, Application of spectral forcing in lattice-Boltzmann simulations of homogeneous turbulence, *Comput. & Fluids* 35 (10) (2006) 1239–1251.
- [22] P. Lammers, K.N. Beronov, R. Volkert, G. Brenner, F. Durst, Lattice BGK direct numerical simulation of fully developed turbulence in incompressible plane channel flow, *Comput. & Fluids* 35 (10) (2006) 1137–1153.
- [23] H. Xu, W. Tao, Y. Zhang, Lattice Boltzmann model for three-dimensional decaying homogeneous isotropic turbulence, *Phys. Lett. A* 373 (15) (2009) 1368–1373.
- [24] Y. Peng, W. Liao, L.S. Luo, L.P. Wang, Comparison of the lattice Boltzmann and pseudo-spectral methods for decaying turbulence: Low-order statistics, *Comput. & Fluids* 39 (4) (2010) 568–591.
- [25] S.S. Chikatamarla, C.E. Frouzakis, I.V. Karlin, A.G. Tomboulides, K.B. Boulouchos, Lattice Boltzmann method for direct numerical simulation of turbulent flows, *J. Fluid Mech.* 656 (2010) 298–308.
- [26] F. Bösch, S.S. Chikatamarla, I.V. Karlin, Entropic multirelaxation lattice Boltzmann models for turbulent flows, *Phys. Rev. E* 92 (4) (2015) 043309.
- [27] Z. Lu, Y. Liao, D. Qian, J.B. McLaughlin, J.J. Derksen, K. Kontomaris, Large eddy simulations of a stirred tank using the lattice Boltzmann method on a nonuniform grid, *J. Comput. Phys.* 181 (2) (2002) 675–704.
- [28] M. Fernandino, K. Beronov, T. Ytrehus, Large eddy simulation of turbulent open duct flow using a lattice Boltzmann approach, *Math. Comput. Simulation* 79 (5) (2009) 1520–1526.
- [29] M. Weickert, G. Teike, O. Schmidt, M. Sommerfeld, Investigation of the LES WALE turbulence model within the lattice Boltzmann framework, *Comput. Math. Appl.* 59 (7) (2010) 2200–2214.
- [30] P. Sagaut, Toward advanced subgrid models for Lattice-Boltzmann-based Large-eddy simulation: theoretical formulations, *Comput. Math. Appl.* 59 (7) (2010) 2194–2199.
- [31] M. Stiebler, M. Krafczyk, S. Freudiger, M. Geier, Lattice Boltzmann large eddy simulation of subcritical flows around a sphere on non-uniform grids, *Comput. Math. Appl.* 61 (12) (2011) 3475–3484.
- [32] S. Jafari, M. Rahnama, Shear-improved Smagorinsky modeling of turbulent channel flow using generalized Lattice Boltzmann equation, *Internat. J. Numer. Methods Fluids* 67 (6) (2011) 700–712.
- [33] G. Eitel-Amor, M. Meinke, W. Schröder, A lattice-Boltzmann method with hierarchically refined meshes, *Comput. & Fluids* 75 (2013) 127–139.
- [34] C.M. Teixeira, Incorporating turbulence models into the lattice-Boltzmann method, *Internat. J. Modern Phys. C* 9 (08) (1998) 1159–1175.
- [35] O. Filippova, S. Succi, F. Mazzocco, C. Arrighetti, G. Bella, D. Hänel, Multiscale lattice Boltzmann schemes with turbulence modeling, *J. Comput. Phys.* 170 (2) (2001) 812–829.
- [36] H. Chen, S. Kandasamy, S. Orszag, R. Shock, S. Succi, V. Yakhot, Extended Boltzmann kinetic equation for turbulent flows, *Science* 301 (5633) (2003) 633–636.
- [37] C. Shu, Y. Peng, C.F. Zhou, Y.T. Chew, Application of Taylor series expansion and Least-squares-based lattice Boltzmann method to simulate turbulent flows, *J. Turbul.* (7) (2006) N38.
- [38] K. Li, C. Zhong, C. Zhuo, J. Cao, Non-body-fitted Cartesian-mesh simulation of highly turbulent flows using multi-relaxation-time lattice Boltzmann method, *Comput. Math. Appl.* 63 (10) (2012) 1481–1496.
- [39] H. Sajjadi, M. Salmanzadeh, G. Ahmadi, S. Jafari, Lattice Boltzmann method and RANS approach for simulation of turbulent flows and particle transport and deposition, *Particuology* 30 (2017) 62–72.
- [40] S. Shu, N. Yang, Numerical study and acceleration of LBM-RANS simulation of turbulent flow, *Chin. J. Chem. Eng.* (2017).
- [41] G. Guzel, I. Koc, Simulation of turbulent flows using a finite-volume based lattice Boltzmann flow solver, *Commun. Comput. Phys.* 17 (01) (2015) 213–232.
- [42] P.L. Bhatnagar, E.P. Gross, M. Krook, A model for collision processes in gases. I. Small amplitude processes in charged and neutral one-component systems, *Phys. Rev.* 94 (3) (1954) 511.
- [43] X. He, L.S. Luo, Lattice Boltzmann model for the incompressible Navier–Stokes equation, *J. Stat. Phys.* 88 (3) (1997) 927–944.
- [44] V. Sofonea, R.F. Sekerka, Viscosity of finite difference lattice Boltzmann models, *J. Comput. Phys.* 184 (2) (2003) 422–434.
- [45] P. Spalart, S. Allmaras, A one-equation turbulence model for aerodynamic flows, in: 30th Aerospace Sciences Meeting and Exhibit, 1992, January, p. 439.
- [46] F.R. Menter, M. Kuntz, R. Langtry, Ten years of industrial experience with the SST turbulence model, *Turbul. Heat Mass Transfer* 4 (1) (2003) 625–632.
- [47] P.R. Spalart, C.L. Rumsey, Effective inflow conditions for turbulence models in aerodynamic calculations, *AIAA J.* 45 (10) (2007) 2544–2553.
- [48] G.S. Jiang, C.W. Shu, Efficient implementation of weighted ENO schemes, *J. Comput. Phys.* 126 (1) (1996) 202–228.
- [49] Christopher Rumsey, <http://turbmodels.larc.nasa.gov/>.
- [50] F.M. White, *Viscous Flow*, Me Graw Hill, 1974.
- [51] N. Gregory, C.L. O'Reilly, Low-speed aerodynamic characteristics of NACA 0012 aerofoil section, including the effects of upper-surface roughness simulating hoar frost (Low speed aerodynamic characteristics of airfoil profiles including effects of upper surface roughness simulating hoar frost), 1970.
- [52] C.L. Ladson, Effects of independent variation of Mach and Reynolds numbers on the low-speed aerodynamic characteristics of the NACA 0012 airfoil section, 1988.